

LOUISA COUNTY AP, VA, USA

WMO: 723114

Lat: 38.010N

Lon: 77.970W

Elev: 150

StdP: 99.53

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.4	-19.0	0.7	-4.7	-16.3	0.9	-3.8	9.1	5.4	8.2	5.7	1.4	280	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	34.7	25.3	33.0	24.8	32.3	24.6	27.5	31.0	26.6	30.6	25.7	30.2	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	22.2	29.2	25.9	21.6	29.0	24.1	19.4	27.6	87.8	30.9	83.7	30.7	80.6	30.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.0	5.1	-14.9	36.2	3.1	1.8	-17.1	37.5	-18.9	38.6	-20.6	39.6	-22.8	40.9		
(5)			-15.2	28.3	3.0	1.2	-17.4	29.1	-19.1	29.8	-20.8	30.5	-22.9	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.9	4.8	9.0	14.8	19.2	23.3	25.4	24.8	21.1	15.0	9.1	5.0
	DBStd	9.17	6.16	5.66	5.86	5.09	4.18	3.21	2.73	2.80	3.48	4.76	5.20	5.20
	HDD10.0	766	234	164	93	13	0	0	0	0	0	11	78	173
	HDD18.3	2161	479	379	293	131	42	3	0	0	14	126	278	415
	CDD10.0	2437	14	18	62	157	284	398	479	457	334	166	52	17
	CDD18.3	791	1	1	3	24	68	151	221	199	98	22	2	1
	CDH23.3	7626	2	6	51	304	661	1474	2237	1881	802	187	21	2
	CDH26.7	3066	0	1	6	101	208	597	1026	806	271	46	3	0
Wind	WSAvg	1.9	2.1	2.3	2.5	2.4	1.8	1.6	1.4	1.3	1.5	1.7	1.9	1.9
Precipitation	PrecAvg	1143	80	71	103	78	107	102	112	93	112	96	87	84
	PrecMax	1704	166	209	226	162	209	187	220	175	259	230	188	183
	PrecMin	935	27	17	5	28	28	26	33	28	16	4	7	38
	PrecStd	185	36	44	50	30	51	46	51	36	72	63	52	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.9	27.5	32.3	32.5	35.3	37.3	36.3	33.8	31.0	26.3	22.3
		MCWB	16.6	16.4	16.9	18.8	23.4	25.6	25.2	26.1	23.7	23.2	18.3	16.5
	2%	DB	18.3	20.1	24.1	29.8	31.1	33.0	35.0	34.0	32.1	27.7	22.6	18.6
		MCWB	14.4	13.9	15.2	18.4	22.8	25.0	25.5	25.3	24.0	20.6	16.9	15.8
	5%	DB	16.1	17.3	22.1	27.3	28.9	32.2	33.0	32.7	29.7	25.6	20.2	16.3
		MCWB	11.6	11.1	14.1	17.7	21.9	24.7	25.1	25.0	22.9	19.5	15.8	12.9
	10%	DB	12.5	14.0	19.0	24.0	27.4	30.3	32.2	31.2	27.7	22.8	17.8	13.6
		MCWB	8.7	9.1	13.2	16.4	21.0	23.8	25.0	24.4	21.9	18.9	13.9	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.8	17.7	19.1	22.2	25.6	28.0	28.4	28.6	27.1	24.3	21.4	19.3
		MCDB	19.7	21.7	23.7	27.9	30.1	31.7	32.2	32.8	29.5	27.9	22.8	20.6
	2%	WB	15.5	15.4	17.4	20.3	24.1	26.7	27.3	27.5	25.2	22.7	18.5	16.5
		MCDB	18.0	18.6	22.4	25.8	28.3	30.6	30.9	31.4	29.4	26.1	21.1	17.8
	5%	WB	12.2	12.5	15.7	19.1	23.0	25.6	26.5	26.6	24.1	21.3	16.7	13.6
		MCDB	14.6	15.8	20.2	24.3	27.8	30.4	31.1	30.1	27.5	23.8	19.3	15.7
	10%	WB	9.1	9.8	13.9	17.7	21.9	24.6	25.7	25.5	23.1	19.6	14.8	10.7
		MCDB	12.3	13.9	18.3	22.3	26.2	29.0	30.6	29.4	26.2	22.4	17.5	12.4
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	12.6	13.1	12.2	11.7	11.4	11.3	11.4	12.3	12.2	10.5
		MCDBR	13.4	15.0	15.6	16.1	14.0	13.4	13.0	12.9	14.0	14.3	14.3	12.7
		MCWBR	9.8	10.0	8.9	7.2	7.1	6.5	5.5	5.4	6.7	7.9	8.9	9.6
	5% WB	MCDBR	11.9	13.4	13.5	13.3	12.0	11.8	11.1	10.7	10.9	11.1	11.5	11.1
		MCWBR	10.1	10.3	9.1	7.0	6.7	6.3	5.7	5.6	6.1	7.0	8.4	9.7
Clear-Sky Solar Irradiance	taub	0.296	0.312	0.347	0.387	0.429	0.479	0.499	0.502	0.407	0.351	0.317	0.301	
	taud	2.523	2.464	2.413	2.320	2.222	2.127	2.095	2.066	2.318	2.470	2.523	2.544	
	Ebn at Noon	895	921	915	892	856	810	790	777	840	863	860	864	
	Edh at Noon	77	93	108	125	141	155	159	160	117	91	77	71	
All-Sky Solar Radiation	RadAvg	2.25	3.03	4.02	5.02	5.64	6.18	5.93	5.33	4.39	3.41	2.52	1.97	
	RadStd	0.18	0.34	0.28	0.38	0.52	0.43	0.32	0.26	0.50	0.39	0.19	0.18	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (72 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	+134	+80

Nomenclature: See separate page